

## AKDENIZ, Cyprus

WMO: 175050

Lat: 35.300N

Lon: 32.967E

Elev: 89

StdP: 100.26

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |            |            | 99.6%                         |            |            | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>5.1 | (c)<br>6.7 | (d)<br>-4.4                   | (e)<br>2.6 | (f)<br>8.2 | (g)<br>-2.2 | (h)<br>3.2 | (i)<br>10.2 | (j)<br>12.2           | (k)<br>12.7 | (l)<br>10.5 | (m)<br>12.3 | (n)<br>3.9            | (o)<br>110 | (p)<br>0.408 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |           |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|-----------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |           |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD      |
| (a)<br>8      | (b)<br>8.9             | (c)<br>36.6     | (d)<br>19.7 | (e)<br>34.8 | (f)<br>20.6 | (g)<br>32.8 | (h)<br>21.9 | (i)<br>26.1         | (j)<br>30.5 | (k)<br>25.6 | (l)<br>30.0 | (m)<br>25.1 | (n)<br>29.5 | (o)<br>4.6           | (p)<br>20 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.9                            | 20.1 | 29.0 | 24.3 | 19.4 | 28.8 | 23.7 | 18.8 | 28.5 | 81.8          | 30.5 | 79.7 | 30.2 | 77.5 | 29.5 | 29.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 10.2          | (b) 8.8 | (c) 7.6 | (e) 2.6                    | (f) 39.7 | (g) 2.2            | (h) 1.4 | (i) 1.0                                            | (j) 40.7 | (k) -0.3   | (l) 41.5 | (m) -1.5   | (n) 42.3 | (o) -3.1   | (p) 43.3 |    |    |
| (a) 10.2          | (b) 8.8 | (c) 7.6 | (e) 0.3                    | (f) 27.2 | (g) 1.8            | (h) 0.9 | (i) -1.0                                           | (j) 27.8 | (k) -2.1   | (l) 28.3 | (m) -3.1   | (n) 28.8 | (o) -4.4   | (p) 29.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 20.1   | 12.2  | 12.9  | 14.5  | 17.6  | 21.5  | 25.2  | 27.9  | 28.4  | 25.7  | 22.2  | 18.1  | 14.0 |
|                                                                                  | DBStd       | 6.10   | 2.26  | 2.34  | 2.19  | 2.30  | 2.57  | 2.53  | 1.97  | 1.58  | 1.64  | 2.07  | 2.13  | 2.45 |
|                                                                                  | HDD10.0     | 18     | 9     | 4     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3    |
|                                                                                  | HDD18.3     | 671    | 190   | 154   | 121   | 40    | 2     | 0     | 0     | 0     | 0     | 1     | 28    | 135  |
|                                                                                  | CDD10.0     | 3694   | 78    | 84    | 141   | 229   | 357   | 456   | 556   | 572   | 472   | 379   | 243   | 127  |
|                                                                                  | CDD18.3     | 1306   | 0     | 1     | 2     | 19    | 101   | 206   | 298   | 313   | 222   | 122   | 21    | 1    |
|                                                                                  | CDH23.3     | 12539  | 0     | 2     | 7     | 151   | 602   | 1771  | 3422  | 3783  | 2028  | 702   | 66    | 4    |
|                                                                                  | CDH26.7     | 4718   | 0     | 0     | 1     | 31    | 183   | 638   | 1449  | 1652  | 623   | 137   | 4     | 1    |
| Wind                                                                             | WSAvg       | 3.3    | 3.3   | 3.5   | 3.6   | 3.3   | 3.3   | 3.4   | 3.2   | 3.1   | 3.4   | 3.0   | 3.0   | 3.2  |
| Precipitation                                                                    | PrecAvg     | 543    | 104   | 91    | 52    | 32    | 25    | 11    | 5     | 5     | 8     | 30    | 62    | 115  |
|                                                                                  | PrecMax     | 963    | 320   | 177   | 124   | 92    | 73    | 68    | 24    | 25    | 37    | 125   | 225   | 333  |
|                                                                                  | PrecMin     | 302    | 14    | 23    | 0     | 1     | 1     | 0     | 0     | 0     | 0     | 1     | 0     | 11   |
|                                                                                  | PrecStd     | 147    | 69    | 39    | 29    | 21    | 18    | 16    | 7     | 7     | 11    | 27    | 47    | 80   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 19.0  | 22.4  | 24.1  | 30.3  | 33.1  | 38.2  | 38.7  | 38.2  | 35.6  | 32.3  | 27.1  | 22.8 |
|                                                                                  |             | MCWB   | 13.6  | 15.0  | 14.9  | 16.1  | 18.1  | 20.1  | 19.6  | 19.9  | 19.4  | 17.8  | 17.5  | 14.8 |
|                                                                                  | 2%          | DB     | 17.7  | 19.6  | 22.0  | 27.3  | 31.0  | 35.1  | 36.6  | 36.4  | 32.4  | 29.7  | 25.1  | 20.4 |
|                                                                                  |             | MCWB   | 12.9  | 14.2  | 14.6  | 15.4  | 18.0  | 19.6  | 19.5  | 20.1  | 21.4  | 18.9  | 17.1  | 14.9 |
|                                                                                  | 5%          | DB     | 17.0  | 18.1  | 20.2  | 25.1  | 28.6  | 32.4  | 34.5  | 34.4  | 30.7  | 27.9  | 23.6  | 19.2 |
|                                                                                  |             | MCWB   | 12.5  | 13.5  | 14.1  | 15.3  | 17.9  | 20.0  | 21.1  | 21.7  | 22.6  | 19.5  | 16.8  | 14.2 |
|                                                                                  | 10%         | DB     | 16.1  | 17.0  | 18.8  | 23.1  | 26.6  | 29.9  | 32.4  | 32.2  | 29.7  | 26.7  | 22.5  | 18.2 |
|                                                                                  |             | MCWB   | 12.0  | 12.8  | 13.8  | 15.6  | 18.1  | 21.2  | 22.9  | 24.0  | 22.7  | 19.3  | 16.2  | 13.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 14.3  | 16.1  | 16.7  | 18.7  | 21.7  | 24.9  | 26.5  | 27.1  | 25.7  | 22.7  | 19.9  | 17.1 |
|                                                                                  |             | MCDB   | 17.4  | 20.5  | 21.3  | 23.8  | 27.1  | 28.7  | 30.8  | 31.5  | 30.4  | 28.0  | 24.7  | 20.8 |
|                                                                                  | 2%          | WB     | 13.6  | 14.9  | 15.9  | 17.9  | 20.7  | 24.1  | 25.9  | 26.4  | 24.8  | 21.7  | 18.7  | 15.8 |
|                                                                                  |             | MCDB   | 16.9  | 18.7  | 20.1  | 22.8  | 25.6  | 28.2  | 30.3  | 30.8  | 29.4  | 26.4  | 22.9  | 19.3 |
|                                                                                  | 5%          | WB     | 13.0  | 13.9  | 15.1  | 17.2  | 20.1  | 23.5  | 25.4  | 25.9  | 24.2  | 21.1  | 17.8  | 14.9 |
|                                                                                  |             | MCDB   | 16.3  | 17.4  | 18.8  | 21.7  | 24.8  | 27.7  | 29.8  | 30.3  | 28.8  | 25.9  | 22.1  | 18.2 |
|                                                                                  | 10%         | WB     | 12.3  | 13.2  | 14.4  | 16.5  | 19.4  | 22.7  | 24.9  | 25.5  | 23.7  | 20.5  | 17.0  | 14.0 |
|                                                                                  |             | MCDB   | 15.6  | 16.8  | 18.0  | 21.0  | 24.1  | 27.0  | 29.3  | 29.9  | 28.2  | 25.2  | 21.5  | 17.5 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.1   | 7.6   | 8.3   | 9.7   | 9.2   | 8.9   | 9.0   | 8.9   | 8.5   | 9.0   | 8.4   | 7.4  |
|                                                                                  |             | MCDBR  | 7.8   | 8.9   | 10.4  | 13.4  | 12.7  | 13.1  | 12.5  | 12.4  | 10.1  | 10.5  | 9.5   | 8.3  |
|                                                                                  |             | MCWBR  | 4.9   | 5.5   | 6.1   | 6.6   | 6.1   | 6.0   | 6.4   | 6.5   | 5.4   | 5.3   | 5.5   | 5.0  |
|                                                                                  | 5% WB       | MCDBR  | 7.3   | 8.6   | 9.0   | 10.8  | 10.4  | 10.0  | 9.6   | 8.9   | 9.0   | 9.1   | 8.8   | 7.7  |
|                                                                                  |             | MCWBR  | 4.8   | 5.5   | 5.9   | 6.1   | 5.8   | 5.4   | 5.6   | 5.3   | 5.4   | 5.1   | 5.5   | 5.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.363  | 0.388 | 0.419 | 0.459 | 0.449 | 0.409 | 0.425 | 0.436 | 0.407 | 0.402 | 0.376 | 0.360 |      |
|                                                                                  | taud        | 2.383  | 2.291 | 2.198 | 2.102 | 2.193 | 2.357 | 2.322 | 2.287 | 2.369 | 2.358 | 2.388 | 2.414 |      |
|                                                                                  | Ebn at Noon | 826    | 844   | 848   | 830   | 842   | 874   | 857   | 842   | 851   | 819   | 804   | 805   |      |
|                                                                                  | Edh at Noon | 94     | 115   | 137   | 158   | 146   | 124   | 128   | 130   | 114   | 106   | 93    | 86    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.45   | 3.38  | 4.83  | 6.05  | 7.02  | 7.84  | 7.62  | 6.81  | 5.79  | 4.26  | 3.03  | 2.27  |      |
|                                                                                  | RadStd      | 0.20   | 0.30  | 0.25  | 0.30  | 0.35  | 0.16  | 0.14  | 0.11  | 0.17  | 0.20  | 0.18  | 0.20  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (13 neighbors) |  | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -80     | +162    | +80     |

Nomenclature: See separate page